# Lab Report: Image classification experiments

## 1 Introduction

In this report we compare several models on standard benchmarks. The learning rate was set to 0.01 with the Adam optimizer. All experiments were run on a single GPU with 16 GB of memory.

## 2 Results

Figure 1 shows the test accuracy by dataset.

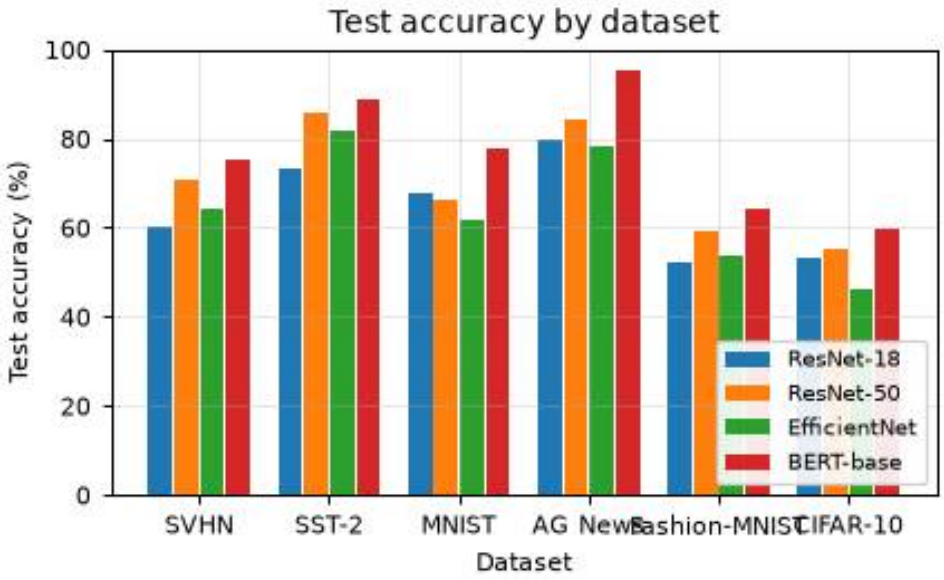



Figure 1: Test accuracy by dataset for the models in this experiment.

On AG News, BERT-base is 20.2 points better than ResNet-18 (Figure 1). As shown in Figure 1, BERT-base reaches 64.2% test accuracy on Fashion-MNIST. As shown in Figure 1, on SST-2, BERT-base is 6.9 points better than EfficientNet.

Figure 2 shows the validation accuracy during training.

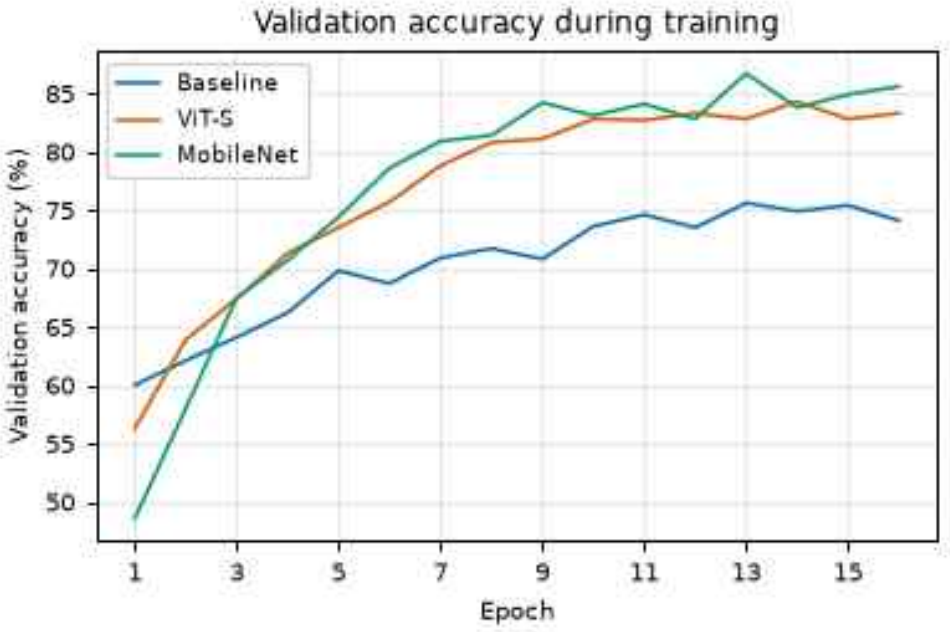



Figure 2: Validation accuracy during training for the models in this experiment.

As shown in Figure 2, at epoch 8, MobileNet achieves 79.3% validation accuracy. As shown in Figure 2, from

epoch 6 to epoch 10, the validation accuracy of Baseline improves by a relative 5.9%. MobileNet reaches 83.1% validation accuracy at epoch 10.

Figure 3 shows the f1 score of ResNet-18 by dataset.

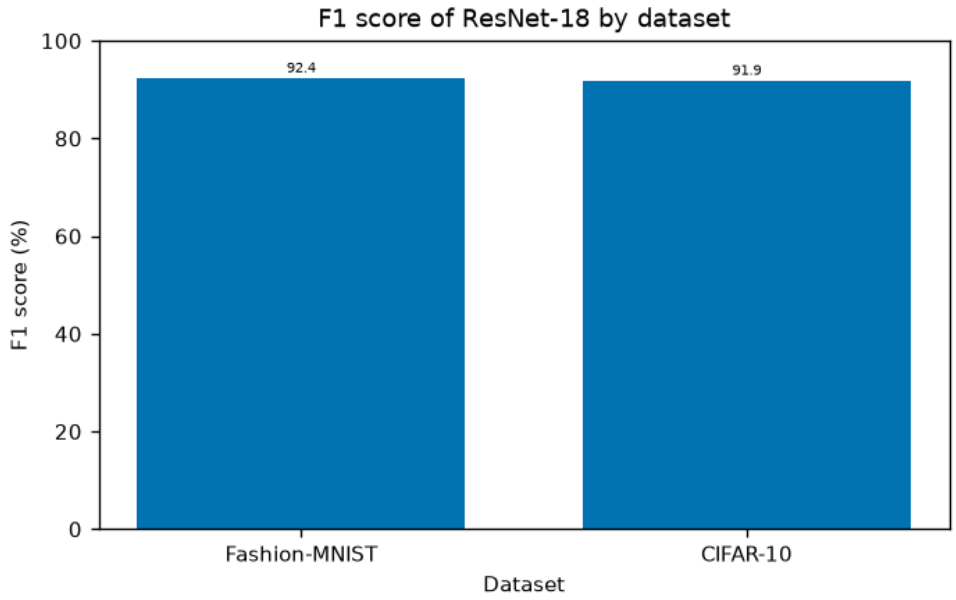



Figure 3: F1 score of ResNet-18 by dataset for the models in this experiment.

ResNet-18 achieves 91.9% F1 score on CIFAR-10 (Figure 3). ResNet-18 reaches 81.9% F1 score on CIFAR-10 (Figure 3).

Figure 4 shows the test accuracy by dataset.

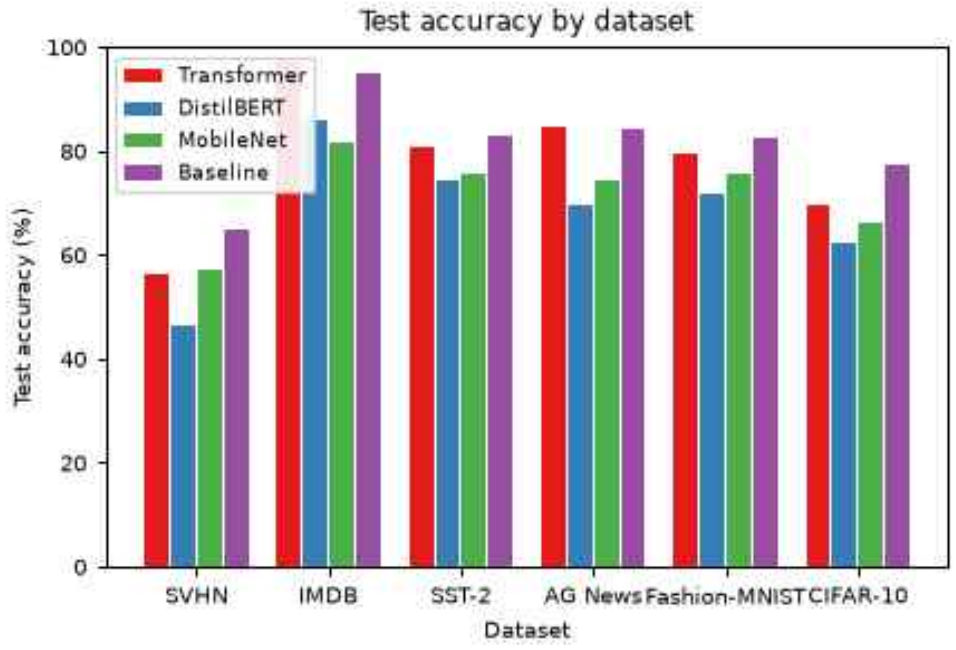



Figure 4: Test accuracy by dataset for the models in this experiment.

DistilBERT reaches 24.5% test accuracy on SST-2. Baseline achieves 32.8% test accuracy on SST-2. On AG News, DistilBERT obtains a test accuracy of 69.5%.

## 3 Discussion

Logistic Regression reaches 72.7% accuracy on IMDB. The differences between the models are discussed in the next section.